

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 6 - Formative Assessment 1

Formative Assessment (FA-1) - Answer Key

Class: Class 6

Assessment Type: Formative Assessment (FA-1)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Internet & ICT Environment, Graphics in ICT

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 — 1 = 5 marks)

Q.No	Answer	Explanation
1	a	ICT stands for Information and Communication Technology
2	b	Google Chrome is a web browser; others are applications
3	a	URL stands for Uniform Resource Locator
4	c	Both Brush and Pencil are used for drawing in MS Paint
5	a	GPS stands for Global Positioning System

Section II: Short Answer Questions (5 — 2 = 10 marks)

- Define ICT and give two examples of ICT devices.**
 - **Answer:** ICT stands for Information and Communication Technology. It refers to technologies used for storing, retrieving, and sending information. Examples: Computer, Smartphone.
 - **Marking:** 1 mark for correct definition, 1 mark for two valid examples.
- What is the difference between hardware and software?**
 - **Answer:** Hardware refers to the physical parts of a computer that you can touch (e.g., keyboard, monitor). Software refers to the programs and instructions that tell the hardware what to do (e.g., MS Word, operating system).
 - **Marking:** 1 mark for defining hardware with example, 1 mark for defining software with example.
- Name any two web browsers available today.**
 - **Answer:** Google Chrome, Mozilla Firefox. (Other valid answers: Opera, Microsoft Edge, Safari)
 - **Marking:** 1 mark for each correct browser.
- What is the use of a search engine? Give one example.**
 - **Answer:** A search engine is used to search for information on the Internet. It helps find websites, images, videos, and other content. Example: Google Search.
 - **Marking:** 1 mark for correct purpose, 1 mark for valid example.
- List any two features of MS Paint.**
 - **Answer:** Features include: Drawing tools (pencil, brush, shapes), Fill with colour tool, Eraser tool, Text tool, Select and crop tool. (Any two)
 - **Marking:** 1 mark for each correct feature.

Section III: Long Answer Questions (5 — 1 = 5 marks)

- Explain the components of a computer system with a neat diagram.**
 - **Answer:** A computer system has three main components: (1) Input devices (keyboard, mouse) receive data. (2) Processing unit (CPU) processes the data. (3) Output devices (monitor, printer)

display results. Storage devices (hard disk, pen drive) store data. A labeled diagram showing all components is expected.

- **Marking:** 1 mark for each component correctly named and explained, 1 mark for diagram, 1 mark for neatness.

2. **What is the Internet? How is it useful in daily life?**

- **Answer:** The Internet is a global network of interconnected computers that allows sharing of information. Uses: Email communication, online education, accessing information, video calling, online banking, shopping.
- **Marking:** 1 mark for definition, 1 mark for listing uses.

Part B: Hands-on Practical Lab Task (5 marks)

1. **Open MS Paint and draw a house with the following elements: (2 marks)**

- **Answer:** Student must draw a house with: triangle roof, rectangular door, square windows, and a tree with circles (leaves) and rectangle (trunk).
- **Marking:** 1 mark for correct shapes and elements, 1 mark for neatness.

2. **Open a web browser and search for “Karnataka state information”. Take a screenshot and save it. (2 marks)**

- **Answer:** Student must open browser, type the search query, take screenshot (PrtScn key), and paste in Paint or save.
- **Marking:** 1 mark for correct search, 1 mark for screenshot saved.

3. **Save your MS Paint drawing with the filename “MyHouse” on the desktop. (1 mark)**

- **Answer:** File > Save As > Desktop > type “MyHouse” > Save.
- **Marking:** 1 mark for correct filename and location.

Part C: Notebook Maintenance (5 marks)

- **Regularity in completing classwork (2 marks):** Check if student has completed all classwork entries.
- **Neatness of work (2 marks):** Overall presentation and handwriting quality.
- **Completion of all activities (1 mark):** All assigned activities present and complete.

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 6 - Formative Assessment 2

Formative Assessment (FA-2) - Answer Key

Class: Class 6

Assessment Type: Formative Assessment (FA-2)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Audio & Visual Communication, Data Representation & Processing

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 — 1 = 5 marks)

Q.No	Answer	Explanation
1	a	.mp3 is an audio file format; others are image, document, spreadsheet
2	a	A microphone is used for recording audio
3	a	LibreOffice Impress is used for creating presentations
4	b	1 Kilobyte (KB) = 1024 bytes
5	c	Line chart is best for showing trends over time

Section II: Short Answer Questions (5 — 2 = 10 marks)

- What is multimedia? Name any two multimedia elements.**
 - **Answer:** Multimedia is the combination of different forms of content such as text, audio, images, animations, and video to convey information. Two elements: Audio, Video.
 - **Marking:** 1 mark for correct definition, 1 mark for two valid elements.
- Define the following terms: (a) Bit (b) Byte**
 - **Answer:** (a) Bit is the smallest unit of data in a computer, represented as 0 or 1. (b) Byte is a group of 8 bits used to represent a single character.
 - **Marking:** 1 mark for correct definition of bit, 1 mark for correct definition of byte.
- What is the difference between a .wav and .mp3 file?**
 - **Answer:** .wav files are uncompressed audio files with higher quality but larger size. .mp3 files are compressed audio files with smaller size but slightly reduced quality.
 - **Marking:** 1 mark for .wav description, 1 mark for .mp3 description.
- Name any two spreadsheet functions used in LibreOffice Calc.**
 - **Answer:** SUM(), AVERAGE(). (Other valid answers: COUNT(), MAX(), MIN())
 - **Marking:** 1 mark for each correct function.
- What are the uses of a video in education?**
 - **Answer:** Videos help in visual learning, demonstrating experiments, explaining complex concepts, virtual field trips, and making lessons more engaging.
 - **Marking:** 1 mark for listing uses, 1 mark for clarity.

Section III: Long Answer Questions (5 — 1 = 5 marks)

- Explain the steps to record audio using Audacity software.**
 - **Answer:** Step 1: Open Audacity. Step 2: Click the red Record button. Step 3: Speak into the microphone. Step 4: Click Stop when done. Step 5: Use Edit > Trim to remove extra silence. Step 6: Go to File > Export Audio to save as .mp3 or .wav.
 - **Marking:** 1 mark for each correct step (up to 5 marks).

2. **What is a spreadsheet? Explain any three features of LibreOffice Calc.**

- **Answer:** A spreadsheet is an application used to organize, analyze, and store data in tabular form. Features: (1) Formulas for calculations (SUM, AVERAGE). (2) Charts for data visualization. (3) Sorting and filtering data.
- **Marking:** 1 mark for definition, 1 mark each for three features.

Part B: Hands-on Practical Lab Task (5 marks)

1. **Open LibreOffice Calc and create a table of marks for 5 subjects with columns: Subject, Marks Obtained, Total Marks. (2 marks)**

- **Answer:** Student must create a table with headers and fill in 5 subjects with marks.
- **Marking:** 1 mark for correct structure, 1 mark for complete data.

2. **Use a formula to calculate the percentage for each subject. (2 marks)**

- **Answer:** Formula: $=(\text{Marks Obtained}/\text{Total Marks}) * 100$. Applied to each row.
- **Marking:** 1 mark for correct formula, 1 mark for all percentages calculated.

3. **Insert a bar chart to display the marks obtained in each subject. (1 mark)**

- **Answer:** Select data > Insert > Chart > Bar chart.
- **Marking:** 1 mark for correct chart insertion.

Part C: Notebook Maintenance (5 marks)

- **Regularity in completing classwork (2 marks):** Check if student has completed all classwork entries.
- **Neatness of work (2 marks):** Overall presentation and handwriting quality.
- **Completion of all activities (1 mark):** All assigned activities present and complete.

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ICT Class 6 - Formative Assessment 3

Formative Assessment (FA-3) - Answer Key

Class: Class 6

Assessment Type: Formative Assessment (FA-3)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Programming - Scratch, Software Applications

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 — 1 = 5 marks)

Q.No	Answer	Explanation
1	b	Scratch is a visual programming language
2	b	The stage is where the output is displayed
3	b	Forever block repeats an action indefinitely
4	a	LibreOffice Writer is used for creating documents
5	a	A database is used to store and retrieve data

Section II: Short Answer Questions (5 — 2 = 10 marks)

- What is a sprite in Scratch? Give one example.**
 - Answer:** A sprite is a character or object on the Scratch stage that can be programmed to perform actions. Example: Cat, Dog, Fish.
 - Marking:** 1 mark for correct definition, 1 mark for valid example.
- Name any three block categories in Scratch.**
 - Answer:** Motion, Looks, Sound. (Other valid answers: Events, Control, Sensing, Operators, Variables)
 - Marking:** 1 mark for each correct category.
- What is the difference between a loop and a condition in Scratch?**
 - Answer:** A loop repeats a set of blocks multiple times (e.g., repeat 10). A condition checks whether a statement is true or false and performs an action accordingly (e.g., if-then).
 - Marking:** 1 mark for explaining loop, 1 mark for explaining condition.
- Define the term “word processing” and name one word processor.**
 - Answer:** Word processing is the creation, editing, formatting, and printing of text documents using a computer. Word processor: LibreOffice Writer or Microsoft Word.
 - Marking:** 1 mark for correct definition, 1 mark for valid word processor.
- What is a utility software? Give two examples.**
 - Answer:** Utility software is system software designed to help maintain, optimize, or manage computer resources. Examples: Antivirus, Disk Cleanup.
 - Marking:** 1 mark for correct definition, 1 mark for two valid examples.

Section III: Long Answer Questions (5 — 1 = 5 marks)

- Explain the Scratch interface with its main components.**
 - Answer:** The Scratch interface has: (1) Stage area where output is displayed. (2) Sprite list showing all sprites. (3) Block palette with categorized code blocks. (4) Script area where blocks are assembled. (5) Menu bar for file operations. (6) Green flag to start and red stop button to end programs.

- **Marking:** 1 mark for each component explained (up to 5 marks).
2. **What are the advantages of using presentation software? Explain any three features of LibreOffice Impress.**
 - **Answer:** Advantages: Organizes information visually, supports multimedia, easy to present to large audiences. Features: (1) Slide transitions and animations. (2) Insert images, charts, and videos. (3) Design templates for consistent formatting.
 - **Marking:** 1 mark for advantages, 1 mark each for three features.

Part B: Hands-on Practical Lab Task (5 marks)

1. **Open Scratch and create a simple animation where a cat moves from one side of the stage to the other. (2 marks)**
 - **Answer:** Use “when flag clicked” > “glide 2 secs to x: 200 y: 0” or use “move 10 steps” in a loop.
 - **Marking:** 1 mark for correct blocks, 1 mark for animation working.
2. **Add a costume change to the sprite so it appears to walk. (2 marks)**
 - **Answer:** Use “next costume” block inside a loop with “wait 0.3 secs” to simulate walking.
 - **Marking:** 1 mark for costume change, 1 mark for proper timing.
3. **Use a “repeat” block to make the sprite move in a square pattern. (1 mark)**
 - **Answer:** Use “repeat 4” containing “move 50 steps” and “turn 90 degrees”.
 - **Marking:** 1 mark for correct repeat and turn combination.

Part C: Notebook Maintenance (5 marks)

- **Regularity in completing classwork (2 marks):** Check if student has completed all classwork entries.
- **Neatness of work (2 marks):** Overall presentation and handwriting quality.
- **Completion of all activities (1 mark):** All assigned activities present and complete.

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ICT Class 6 - Formative Assessment 4

Formative Assessment (FA-4) - Answer Key

Class: Class 6

Assessment Type: Formative Assessment (FA-4)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Programming - Scratch, Software Applications, Revision

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 — 1 = 5 marks)

Q.No	Answer	Explanation
1	b	Motion category contains “move” and “turn” blocks
2	b	The Scratch project file extension is .sb3
3	a	The intersection of a row and column is called a cell
4	c	Keyboard is an input device; monitor, speaker, printer are output
5	c	Firewall is a type of security software

Section II: Short Answer Questions (5 — 2 = 10 marks)

- What is the difference between a variable and a list in Scratch?**
 - **Answer:** A variable stores a single value (e.g., score = 10). A list stores multiple values in an ordered collection (e.g., scores = [10, 20, 30]).
 - **Marking:** 1 mark for explaining variable, 1 mark for explaining list.
- Name any two keyboard shortcuts used in LibreOffice Writer.**
 - **Answer:** Ctrl + C (Copy), Ctrl + V (Paste). (Other valid answers: Ctrl + B (Bold), Ctrl + S (Save), Ctrl + Z (Undo))
 - **Marking:** 1 mark for each correct shortcut with its function.
- What is cloud computing? Give one example.**
 - **Answer:** Cloud computing is storing and accessing data and programs over the Internet instead of a local hard drive. Example: Google Drive, Dropbox.
 - **Marking:** 1 mark for correct definition, 1 mark for valid example.
- Explain the purpose of a backup. Why is it important?**
 - **Answer:** A backup is a copy of data stored separately to prevent loss. It is important because data can be lost due to hardware failure, viruses, or accidental deletion. Backup ensures data can be recovered.
 - **Marking:** 1 mark for defining backup, 1 mark for explaining importance.
- What is the difference between RAM and ROM?**
 - **Answer:** RAM (Random Access Memory) is temporary memory that stores data currently in use and is lost when the computer is turned off. ROM (Read Only Memory) is permanent memory that stores instructions and cannot be modified easily.
 - **Marking:** 1 mark for RAM explanation, 1 mark for ROM explanation.

Section III: Long Answer Questions (5 — 1 = 5 marks)

- Write a short note on the uses of computers in education. Give at least three examples.**
 - **Answer:** Computers are widely used in education for: (1) Online learning through video lectures and e-books. (2) Research using search engines and digital libraries. (3) Creating

presentations and projects. (4) Virtual labs for science experiments. (5) Language learning through apps.

- **Marking:** 1 mark for introduction, 1 mark each for three well-explained uses.

2. **Explain any four basic operations of a computer with examples.**

- **Answer:** (1) Input: Entering data using keyboard or mouse. (2) Processing: CPU performs calculations and operations on data. (3) Output: Displaying results on monitor or printing. (4) Storage: Saving data on hard disk or pen drive for later use.
- **Marking:** 1 mark for each operation correctly explained with example.

Part B: Hands-on Practical Lab Task (5 marks)

1. **Open LibreOffice Writer and type a paragraph of 5 sentences about “My Favourite Subject”. Apply formatting: (2 marks)**

- **Answer:** Title should be bold and size 16. Body text should be size 12. Student must type 5 meaningful sentences.
- **Marking:** 1 mark for correct formatting, 1 mark for complete paragraph.

2. **Open Scratch and create a project with two sprites interacting with each other using “say” blocks. (2 marks)**

- **Answer:** Two sprites with “say” blocks taking turns (e.g., Cat says “Hello!”, Fish says “Hi!”).
- **Marking:** 1 mark for two sprites, 1 mark for interaction using say blocks.

3. **Save both files on the desktop with appropriate filenames. (1 mark)**

- **Answer:** Writer file saved as “MyFavouriteSubject.odt” and Scratch project saved with a descriptive name.
- **Marking:** 1 mark for both files saved correctly.

Part C: Notebook Maintenance (5 marks)

- **Regularity in completing classwork (2 marks):** Check if student has completed all classwork entries.
- **Neatness of work (2 marks):** Overall presentation and handwriting quality.
- **Completion of all activities (1 mark):** All assigned activities present and complete.

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ICT Class 6 - Practical Lab Exam 1

Practical Lab Exam 1 - Answer Key

Class: Class 6

Assessment Type: Practical Lab Exam 1

Total Marks: 20

Duration: 45 minutes

Topics Covered: Practical Activities: Ch 1-4

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Lab Task 1 (5 marks)

Open MS Paint and complete the following:

- Draw a scenery with the following elements: (3 marks)**
 - **Answer:** Student must draw: Sun using circle and lines for rays. Mountains using triangles. River using curved lines. Two trees using circles for leaves and rectangles for trunks. House using rectangle with triangle roof.
 - **Marking:** 1 mark for sun, 1 mark for mountains and river, 1 mark for trees and house.
- Use the “Fill with colour” tool to colour the drawing with at least 4 different colours. (1 mark)**
 - **Answer:** Student must use the bucket tool to fill at least 4 different areas with different colours.
 - **Marking:** 1 mark for using fill tool with 4+ colours.
- Save the file with the name “MyScenery” on the desktop. (1 mark)**
 - **Answer:** File > Save As > Desktop > type “MyScenery” > Save.
 - **Marking:** 1 mark for correct filename and location.

Lab Task 2 (5 marks)

Open a web browser and complete the following:

- Open Google and search for “facts about Karnataka”. (1 mark)**
 - **Answer:** Student must open browser, go to google.com, and type the search query.
 - **Marking:** 1 mark for correct search.
- Open Google Earth and navigate to Bengaluru city. Take a screenshot. (2 marks)**
 - **Answer:** Student must open Google Earth, search for Bengaluru, navigate to the city, and take a screenshot using PrtScn or the screenshot tool.
 - **Marking:** 1 mark for navigating to Bengaluru, 1 mark for taking screenshot.
- Open a new tab and visit any educational website. Write the name of the website and one thing you learned from it. (2 marks)**
 - **Answer:** Student must visit an educational website (e.g., Khan Academy, Wikipedia, Byju’s) and write the name and one learning point.
 - **Marking:** 1 mark for visiting website, 1 mark for writing name and learning.

Lab Task 3 (5 marks)

Open LibreOffice Calc and complete the following:

- Create a table with the following data: (2 marks)**

Subject	Marks Obtained	Total Marks	Percentage
Mathematics	85	100	

Science	78	100	
English	92	100	
Kannada	88	100	
Social Science	75	100	

- **Answer:** Student must enter all data in correct columns with proper headers.
 - **Marking:** 1 mark for correct headers, 1 mark for complete data entry.
2. **Use a formula to calculate the percentage for each subject. (2 marks)**
 - **Answer:** Formula: $= (B2/C2) * 100$ applied to each row. Results: 85%, 78%, 92%, 88%, 75%.
 - **Marking:** 1 mark for correct formula, 1 mark for all percentages calculated.
 3. **Calculate the total and average marks using formulas. (1 mark)**
 - **Answer:** Total: $= \text{SUM}(B2:B6) = 418$. Average: $= \text{AVERAGE}(B2:B6) = 83.6$.
 - **Marking:** 1 mark for correct total and average formulas.

Viva Voce (5 marks)

Answer the following questions orally:

1. **What does ICT stand for? (1 mark)**
 - **Answer:** Information and Communication Technology.
2. **Name any two input devices and two output devices. (1 mark)**
 - **Answer:** Input: Keyboard, Mouse. Output: Monitor, Printer.
3. **What is the difference between hardware and software? (1 mark)**
 - **Answer:** Hardware is the physical part of a computer; software is the set of instructions that tells the hardware what to do.
4. **What is the use of a search engine? (1 mark)**
 - **Answer:** A search engine helps find information on the Internet.
5. **What file extension is used for spreadsheet files in LibreOffice? (1 mark)**
 - **Answer:** .ods (OpenDocument Spreadsheet).

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ICT Class 6 - Practical Lab Exam 2

Practical Lab Exam 2 - Answer Key

Class: Class 6

Assessment Type: Practical Lab Exam 2

Total Marks: 20

Duration: 45 minutes

Topics Covered: Practical Activities: Ch 1-6

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Lab Task 1 (5 marks)

Open Scratch and create a project with the following:

1. **Add two sprites (a cat and a fish) to the stage. (1 mark)**
 - **Answer:** Click “Choose a Sprite” button and add Cat and Fish sprites from the library.
 - **Marking:** 1 mark for adding both sprites.
2. **Write a script so the cat says “Hello! I am a cat” when the green flag is clicked. (2 marks)**
 - **Answer:** Use “when flag clicked” block > attach “say [Hello! I am a cat] for (2) secs” block.
 - **Marking:** 1 mark for “when flag clicked” block, 1 mark for “say” block.
3. **Write a script so the fish moves continuously from left to right on the stage. (1 mark)**
 - **Answer:** Use “when flag clicked” > “forever” > “change x by (5)” > “if on edge, bounce”.
 - **Marking:** 1 mark for correct movement script.
4. **Add a backdrop to the stage. (1 mark)**
 - **Answer:** Click “Choose a Backdrop” button and select a backdrop from the library.
 - **Marking:** 1 mark for adding a backdrop.

Lab Task 2 (5 marks)

Open LibreOffice Writer and complete the following:

1. **Type a paragraph of at least 5 sentences about “My School”. (1 mark)**
 - **Answer:** Student must type 5 meaningful sentences about their school.
 - **Marking:** 1 mark for complete paragraph.
2. **Apply the following formatting: (2 marks)**
 - **Answer:** Title: Select title text > Bold (Ctrl+B) > Font size 16 > Center alignment. Heading: Select heading > Bold > Font size 14. Body: Select body text > Font size 12 > Justified alignment.
 - **Marking:** 1 mark for title formatting, 1 mark for heading and body formatting.
3. **Insert a table with 3 columns and 4 rows showing your weekly timetable. (1 mark)**
 - **Answer:** Table > Insert Table > 3 columns, 4 rows > Fill with timetable data.
 - **Marking:** 1 mark for correct table insertion.
4. **Save the file as “MySchool” on the desktop. (1 mark)**
 - **Answer:** File > Save As > Desktop > type “MySchool” > Save.
 - **Marking:** 1 mark for correct filename and location.

Lab Task 3 (5 marks)

Open LibreOffice Impress and complete the following:

1. **Create a new presentation with 3 slides. (1 mark)**
 - **Answer:** File > New Presentation > Add slides until there are 3.
 - **Marking:** 1 mark for 3 slides created.

2. **Slide 1: Title slide with the title “My Favourite Book” and your name. (1 mark)**
 - **Answer:** Type “My Favourite Book” as title and student’s name as subtitle.
 - **Marking:** 1 mark for correct title and name.
3. **Slide 2: Add the name of the book, author, and a brief summary (3-4 lines). (1 mark)**
 - **Answer:** Student must enter book details and a short summary.
 - **Marking:** 1 mark for complete content.
4. **Slide 3: Add 3 reasons why you liked the book. (1 mark)**
 - **Answer:** Student must list 3 valid reasons in bullet points.
 - **Marking:** 1 mark for 3 reasons.
5. **Apply a design template to the presentation. (1 mark)**
 - **Answer:** Go to Slide > Slide Design > Select a template > Apply to all slides.
 - **Marking:** 1 mark for applying template.

Viva Voce (5 marks)

Answer the following questions orally:

1. **What is Scratch? Who developed it? (1 mark)**
 - **Answer:** Scratch is a visual programming language. It was developed by MIT Media Lab.
2. **Name any three block categories in Scratch. (1 mark)**
 - **Answer:** Motion, Looks, Sound. (Other valid answers: Events, Control, Sensing)
3. **What is the difference between a word processor and a presentation software? (1 mark)**
 - **Answer:** A word processor creates text documents; presentation software creates slide-based visual presentations.
4. **What is a variable? Give one example from Scratch. (1 mark)**
 - **Answer:** A variable stores a value. Example: “score” variable in a game.
5. **Why do we use utility software on a computer? Give one example. (1 mark)**
 - **Answer:** Utility software helps maintain and manage the computer. Example: Antivirus software.

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ICT Class 6 - Summative Assessment 1

Summative Assessment (SA-1) - Answer Key

Class: Class 6

Assessment Type: Summative Assessment (SA-1)

Total Marks: 40

Duration: 2 hours

Topics Covered: Ch 1-4: Internet, Graphics, Audio & Visual, Data Representation

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Multiple Choice Questions (10 – 1 = 10 marks)

Q.No	Answer	Explanation
1	a	ICT stands for Information and Communication Technology
2	b	Google Search is a search engine; Chrome is a browser
3	c	Ctrl + C is the shortcut for copying text
4	c	Monitor is an output device
5	b	1 Megabyte (MB) = 1024 Kilobytes (KB)
6	b	Select tool is used for selecting a portion of an image
7	b	Microphone converts sound waves to electrical signals
8	b	LibreOffice Calc is used for creating spreadsheets
9	a	1010 is a binary number (contains only 0 and 1)
10	b	Pie chart shows parts of a whole

Part B: Short Answer Questions (8 – 2 = 16 marks)

- What is the difference between hardware and software? Give two examples of each.**
 - **Answer:** Hardware refers to physical components you can touch. Examples: Keyboard, Monitor. Software refers to programs and instructions. Examples: MS Word, Operating System.
 - **Marking:** 1 mark for hardware definition and examples, 1 mark for software definition and examples.
- Explain the difference between a web browser and a search engine.**
 - **Answer:** A web browser is software used to access and view websites (e.g., Google Chrome, Firefox). A search engine is a website that helps find other websites by searching the Internet (e.g., Google Search).
 - **Marking:** 1 mark for browser explanation, 1 mark for search engine explanation.
- What is image editing? Name any two image editing tools available in MS Paint.**
 - **Answer:** Image editing is modifying images to improve quality or create new designs. Two tools in MS Paint: Select tool, Fill with colour tool.
 - **Marking:** 1 mark for definition, 1 mark for correct tools.
- Define the terms: (a) Bit (b) Byte (c) Kilobyte**
 - **Answer:** (a) Bit is the smallest unit of data (0 or 1). (b) Byte = 8 bits. (c) Kilobyte (KB) = 1024 bytes.
 - **Marking:** 1 mark for bit and byte, 1 mark for kilobyte.
- What is multimedia? Name its five components.**
 - **Answer:** Multimedia is the combination of multiple forms of content. Five components: Text, Audio, Video, Graphics, Animation.
 - **Marking:** 1 mark for definition, 1 mark for listing five components.

6. **What is the difference between RAM and ROM?**

- **Answer:** RAM is temporary memory that loses data when power is turned off. ROM is permanent memory that retains data and cannot be easily modified.
- **Marking:** 1 mark for RAM, 1 mark for ROM.

7. **Explain any two uses of the Internet in education.**

- **Answer:** (1) Online learning: Students can attend video lectures and access e-books. (2) Research: Students can search for information using search engines and digital libraries.
- **Marking:** 1 mark for each well-explained use.

8. **What is a file extension? Name extensions for three different file types.**

- **Answer:** A file extension is a suffix at the end of a filename that indicates the file format. Three extensions: .odt (document), .ods (spreadsheet), .jpg (image).
- **Marking:** 1 mark for definition, 1 mark for three correct extensions.

Part C: Long Answer Questions (4 — 3 = 12 marks)

1. **Explain the components of a computer system with a neat labeled diagram.**

- **Answer:** A computer system has: (1) Input devices (keyboard, mouse) for entering data. (2) CPU (Central Processing Unit) for processing. (3) Output devices (monitor, printer) for displaying results. (4) Storage devices (hard disk, pen drive) for saving data. (5) Memory (RAM, ROM). Diagram showing all components with labels is required.
- **Marking:** 1 mark each for 5 components, 1 mark for diagram, 1 mark for neatness (3 marks total).

2. **What is the Internet? Explain any four services available on the Internet.**

- **Answer:** The Internet is a global network of interconnected computers. Four services: (1) Email for sending electronic messages. (2) World Wide Web for accessing websites. (3) FTP for transferring files. (4) Video conferencing for face-to-face communication online.
- **Marking:** 1 mark for definition, 1 mark each for four services.

3. **Describe the steps to create a simple drawing in MS Paint. Mention any five tools used.**

- **Answer:** Steps: Open MS Paint > Select tool > Draw on canvas > Colour the drawing > Save the file. Five tools: Pencil, Brush, Shapes (Rectangle, Circle), Fill with colour, Text tool.
- **Marking:** 1 mark for steps, 1 mark each for five tools.

4. **What is a spreadsheet? Explain any three functions used in LibreOffice Calc with examples.**

- **Answer:** A spreadsheet organizes data in rows and columns for calculations. Three functions: (1) SUM: =SUM(A1:A5) adds values. (2) AVERAGE: =AVERAGE(A1:A5) calculates mean. (3) MAX: =MAX(A1:A5) finds the highest value.
- **Marking:** 1 mark for definition, 1 mark each for three functions with examples.

Part D: Application Based Questions (2 — 1 = 2 marks)

1. **Ravi wants to send a letter to his teacher using a computer. Which software should he use and why?**

- **Answer:** Ravi should use a word processor like LibreOffice Writer because it allows formatting text, adding margins, and printing professional-looking letters.
- **Marking:** 1 mark for correct software with reason.

2. **A school wants to display the results of all students in a bar graph. Which software can be used and what are the steps involved?**

- **Answer:** LibreOffice Calc can be used. Steps: Enter student data in columns > Select the data > Insert > Chart > Choose Bar Chart > Customize and insert.
- **Marking:** 1 mark for correct software and steps.

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ICT Class 6 - Summative Assessment 2

Summative Assessment (SA-2) - Answer Key

Class: Class 6

Assessment Type: Summative Assessment (SA-2)

Total Marks: 40

Duration: 2 hours

Topics Covered: Ch 1-6: Full Syllabus - Annual Examination

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Multiple Choice Questions (10 – 1 = 10 marks)

Q.No	Answer	Explanation
1	b	Scratch is a programming language; others are applications
2	b	The stage is where output is displayed in Scratch
3	a	URL stands for Uniform Resource Locator
4	a	Antivirus is a utility software
5	a	A variable stores a value in Scratch
6	a	LibreOffice Impress is used for creating presentations
7	b	Looks category contains “say” and “think” blocks
8	a	The intersection of a row and column is a cell
9	c	Mouse is an input device
10	a	Cloud computing means storing data on remote servers accessed via Internet

Part B: Short Answer Questions (8 – 2 = 16 marks)

- What is the difference between a loop and a condition in Scratch? Give one example of each.**
 - **Answer:** A loop repeats blocks multiple times (e.g., “repeat 10”). A condition checks if something is true and performs an action (e.g., “if touching edge, then bounce”).
 - **Marking:** 1 mark for loop with example, 1 mark for condition with example.
- Explain the difference between hardware and software with examples.**
 - **Answer:** Hardware is the physical part of a computer (e.g., keyboard, monitor). Software is the set of instructions that tells hardware what to do (e.g., LibreOffice Writer, Scratch).
 - **Marking:** 1 mark for hardware with examples, 1 mark for software with examples.
- What is the use of Google Earth? Name any two features.**
 - **Answer:** Google Earth is used for exploring the world using satellite imagery. Two features: (1) View satellite images of any location. (2) Navigate to places and measure distances.
 - **Marking:** 1 mark for use, 1 mark for two features.
- Define the term “multimedia”. List its components.**
 - **Answer:** Multimedia is the use of multiple forms of content to convey information.
Components: Text, Audio, Video, Graphics, Animation.
 - **Marking:** 1 mark for definition, 1 mark for listing components.
- What is a word processor? Name any three formatting options available in LibreOffice Writer.**
 - **Answer:** A word processor is software for creating and editing text documents. Three formatting options: Bold, Italic, Font size.
 - **Marking:** 1 mark for definition, 1 mark for three formatting options.

6. **Explain the purpose of a database. Give one real-life example.**

- **Answer:** A database is used to store, organize, and retrieve large amounts of data efficiently.
Example: School student records database containing names, roll numbers, and marks.
- **Marking:** 1 mark for purpose, 1 mark for example.

7. **What is the difference between .jpg and .png image file formats?**

- **Answer:** .jpg uses lossy compression with smaller file size but loses some quality. .png uses lossless compression with larger file size but retains full quality and supports transparency.
- **Marking:** 1 mark for .jpg, 1 mark for .png.

8. **Why is it important to back up data? Name two methods of backing up data.**

- **Answer:** Backup is important to prevent data loss due to hardware failure, viruses, or accidents. Two methods: (1) Copying to a pen drive. (2) Uploading to cloud storage like Google Drive.
- **Marking:** 1 mark for importance, 1 mark for two methods.

Part C: Long Answer Questions (4 — 3 = 12 marks)

1. **Explain the Scratch interface with its main components. How do you create a simple animation in Scratch?**

- **Answer:** Scratch interface has: (1) Stage for output. (2) Sprite list. (3) Block palette with categories. (4) Script area. (5) Menu bar. To animate: Add sprite > use “when flag clicked” > add “move” and “turn” blocks > use “repeat” for continuous motion > add “next costume” for walking effect.
- **Marking:** 1 mark for each component (3 marks total), 1 mark for animation steps.

2. **What is the Internet? Explain any four uses of the Internet in daily life.**

- **Answer:** The Internet is a global network of computers. Four uses: (1) Email communication. (2) Online education and research. (3) Video calling. (4) Online banking and shopping.
- **Marking:** 1 mark for definition, 1 mark each for four uses.

3. **Describe the steps to create a presentation in LibreOffice Impress. Mention any five features used.**

- **Answer:** Steps: Open Impress > Choose template > Add title and content > Insert images > Add transitions > Save. Five features: Slide transitions, animations, design templates, speaker notes, export to PDF.
- **Marking:** 1 mark for steps, 1 mark each for five features.

4. **Explain the binary number system. Convert the decimal number 25 to binary and vice versa.**

- **Answer:** Binary uses only 0 and 1. Each digit represents a power of 2. Decimal 25 to binary: $25 = 16 + 8 + 1 = 11001$. Binary 11001 to decimal: $16 + 8 + 1 = 25$.
- **Marking:** 1 mark for binary explanation, 1 mark for 25 to binary, 1 mark for binary to decimal.

Part D: Application Based Questions (2 — 1 = 2 marks)

1. **Priya wants to create a quiz project in Scratch where the player answers questions and gets a score. Which Scratch blocks would she use to keep track of the score? Explain.**

- **Answer:** Priya should use a variable called “score”. She would use “set score to 0” at start, “change score by 1” when the answer is correct, and “say” blocks to show the current score.
- **Marking:** 1 mark for using variable with blocks explained.

2. **A shopkeeper wants to maintain records of items sold, their prices, and daily sales. Which software application would you recommend and why?**

- **Answer:** LibreOffice Calc (spreadsheet) is recommended because it can organize data in rows and columns, perform calculations using formulas, create charts for sales analysis, and sort/filter data easily.
- **Marking:** 1 mark for correct software with reasons.

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ICT Class 6 - Unit Test 1

Unit Test 1 - Answer Key

Class: Class 6

Assessment Type: Unit Test 1

Total Marks: 10

Duration: 30 minutes

Topics Covered: Internet & ICT Environment

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: Multiple Choice Questions (5 – 1 = 5 marks)

Q.No	Answer	Explanation
1	a	ICT stands for Information and Communication Technology
2	c	Microsoft Word is not a web browser
3	a	HTTP stands for HyperText Transfer Protocol
4	b	Google Earth is used for viewing maps and satellite imagery
5	d	All options are advantages of the Internet

Section II: Short Answer Questions (5 – 1 = 5 marks)

- What is the difference between the Internet and an intranet?**
 - **Answer:** The Internet is a worldwide network of computers accessible to everyone. An intranet is a private network accessible only within an organization.
 - **Marking:** 1 mark for correct explanation.
- Name any two services available on the Internet.**
 - **Answer:** Email, World Wide Web. (Other valid answers: FTP, video conferencing, online banking)
 - **Marking:** 1 mark for each correct service.
- What is a web address? Give one example.**
 - **Answer:** A web address (URL) is the location of a website on the Internet. Example: www.google.com.
 - **Marking:** 1 mark for definition with example.
- List two safety rules for using the Internet.**
 - **Answer:** (1) Never share personal information with strangers online. (2) Do not open unknown links or email attachments.
 - **Marking:** 1 mark for each valid safety rule.
- What is the use of GPS in daily life? Give one example.**
 - **Answer:** GPS is used for navigation and finding locations. Example: Google Maps for finding directions to a place.
 - **Marking:** 1 mark for correct use with example.

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ICT Class 6 - Unit Test 2

Unit Test 2 - Answer Key

Class: Class 6

Assessment Type: Unit Test 2

Total Marks: 10

Duration: 30 minutes

Topics Covered: Audio & Visual Communication

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: Multiple Choice Questions (5 – 1 = 5 marks)

Q.No	Answer	Explanation
1	a	Microphone is used for recording sound
2	a	.mp4 is a common video file extension
3	c	Multimedia means combination of text, audio, video, and graphics
4	a	GIMP is used for editing images
5	a	A presentation slide can contain text and images

Section II: Short Answer Questions (5 – 1 = 5 marks)

- What is the difference between audio and video?**
 - **Answer:** Audio refers to sound or voice that can be heard. Video refers to moving images with sound that can be seen and heard.
 - **Marking:** 1 mark for correct explanation.
- Name any two multimedia software used in schools.**
 - **Answer:** Audacity (audio editing), GIMP (image editing). (Other valid answers: VLC media player, Blender)
 - **Marking:** 1 mark for each correct software.
- What is a file format? Give two examples of image file formats.**
 - **Answer:** A file format is the way data is organized and stored in a file. Two image formats: .jpg, .png.
 - **Marking:** 1 mark for definition, 1 mark for correct examples.
- Why do we use headphones while listening to audio in a computer lab?**
 - **Answer:** Headphones prevent sound from disturbing others in the lab. They also provide better audio clarity for the user.
 - **Marking:** 1 mark for correct reason.
- What is an infographic? How is it useful?**
 - **Answer:** An infographic is a visual representation of information or data using images, charts, and text. It is useful for presenting complex information in a simple and engaging way.
 - **Marking:** 1 mark for correct explanation.

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ICT Class 6 - Unit Test 3

Unit Test 3 - Answer Key

Class: Class 6

Assessment Type: Unit Test 3

Total Marks: 10

Duration: 30 minutes

Topics Covered: Programming - Scratch

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: Multiple Choice Questions (5 – 1 = 5 marks)

Q.No	Answer	Explanation
1	b	The green flag is used to start the program in Scratch
2	a	A variable in Scratch is used to store a value
3	b	Forever block repeats an action forever
4	a	The “say” block displays a speech bubble
5	a	A sprite is a character or object on the stage

Section II: Short Answer Questions (5 – 1 = 5 marks)

- What is the Scratch stage? What are its dimensions?**
 - **Answer:** The Scratch stage is the area where the output of a Scratch project is displayed. Its dimensions are 480 pixels wide and 360 pixels tall.
 - **Marking:** 1 mark for correct explanation with dimensions.
- Name any three categories of blocks available in Scratch.**
 - **Answer:** Motion, Looks, Sound. (Other valid answers: Events, Control, Sensing, Operators, Variables)
 - **Marking:** 1 mark for each correct category.
- What is a costume in Scratch? Why do we use multiple costumes?**
 - **Answer:** A costume is the appearance or image of a sprite. Multiple costumes are used to create animation effects, such as making a sprite appear to walk or change expressions.
 - **Marking:** 1 mark for defining costume, 1 mark for explaining use of multiple costumes.
- How do you add a new sprite to a Scratch project?**
 - **Answer:** Click on the “Choose a Sprite” button (cat icon) below the stage. You can choose from the library, paint your own, or upload from computer.
 - **Marking:** 1 mark for correct method.
- What is the difference between the “repeat” and “repeat until” blocks?**
 - **Answer:** “Repeat” block repeats a set of blocks a fixed number of times (e.g., repeat 10). “Repeat until” block repeats until a specified condition becomes true.
 - **Marking:** 1 mark for correct explanation of both blocks.

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ICT Class 6 - Unit Test 4

Unit Test 4 - Answer Key

Class: Class 6

Assessment Type: Unit Test 4

Total Marks: 10

Duration: 30 minutes

Topics Covered: Software Applications

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: Multiple Choice Questions (5 – 1 = 5 marks)

Q.No	Answer	Explanation
1	a	LibreOffice Writer is used for creating text documents
2	a	The extension of a Writer document is .odt
3	a	A database is a collection of organized data
4	a	Antivirus is a utility software
5	a	Presentation software is used for creating slideshows

Section II: Short Answer Questions (5 – 1 = 5 marks)

- What is the difference between a word processor and a text editor?**
 - **Answer:** A word processor (e.g., LibreOffice Writer) provides advanced formatting options like fonts, styles, and page layout. A text editor (e.g., Notepad) is basic and only handles plain text without formatting.
 - **Marking:** 1 mark for correct explanation.
- Name any three features of LibreOffice Impress.**
 - **Answer:** (1) Slide transitions and animations. (2) Insert images, charts, and videos. (3) Design templates. (Other valid answers: Speaker notes, handout printing)
 - **Marking:** 1 mark for each correct feature.
- What is a spreadsheet? Name the rows and columns in a spreadsheet.**
 - **Answer:** A spreadsheet is an application used to organize data in rows and columns for calculations and analysis. Rows are numbered (1, 2, 3...) and columns are labeled (A, B, C...).
 - **Marking:** 1 mark for definition, 1 mark for naming rows and columns.
- Why do we use antivirus software on a computer?**
 - **Answer:** Antivirus software is used to detect, prevent, and remove malicious software (malware) such as viruses, worms, and trojans that can damage data and harm the computer.
 - **Marking:** 1 mark for correct explanation.
- What is the purpose of a printer driver?**
 - **Answer:** A printer driver is a software program that allows the computer to communicate with and control the printer. It converts document data into a format the printer can understand.
 - **Marking:** 1 mark for correct explanation.